



HOUSEHOLD AI / SOVEREIGNTY BRIEF

Your home should own the rules, not every model.

Sovereign AI at human scale: an architecture for enforceable household control, portable intelligence, bounded action, continuity, and evidence.

THE ARGUMENT

Sovereignty is an executable property, not a residency promise. The household should control the context, authority, action, evidence, and exit path while models remain replaceable.

WHAT CHANGED IN THIS EDITION

The capability contract proposed across every prior edition has been built and is machine-checkable, and the substitution metric has been computed for the first time — the first measured figure in this work. It found a binding that is better at the job and should still not ship. Earlier this edition: two European obligations in force, model substitution narrowed to what it does not carry, and a sixth finding for the household member who is overruled.

Research synthesis and design hypothesis / Third edition / September 2026. This brief does not claim an implemented runtime or original empirical validation.

FIELD CONTEXT

Sovereignty is decision capacity.

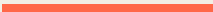
Leading 2025-2026 work treats AI sovereignty as the capacity to make deliberate decisions across infrastructure, data, models, applications, talent, operations, security, and supply dependencies - not as a binary claim that everything is locally owned. [1][2][3]

Full-stack self-sufficiency is unrealistic for most nations and even less practical for a household. The credible alternative is managed interdependence: map dependencies, diversify suppliers, preserve portability, and keep decision rights and continuity under the user's control. [3]

At household scale, sovereignty becomes concrete: who may use which context, where a capability may run, which provider may supply it, what action it may take, and what evidence it must leave behind. Local execution is one instrument of control, not its definition.

Field-level control	Household question	Executable proof
Data and AI	Who may use which context, for what purpose?	Scoped grants, field rules, provenance, deletion and access evidence
Operational	Who can run, support, recover, and change the system?	Safe degraded mode, cached policy, tested recovery and provider exit
Technology and supply	Can models, devices, and services be substituted?	Stable capability contracts, open interfaces, export and regression tests
Security and governance	Who may approve or cause a real-world change?	Typed actions, contextual guards, human checkpoints, receipts

The household should own the rules, not every model.

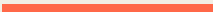


SIX FINDINGS

The control point moves above the model.

Revised in this edition: finding three has partly landed in platform software, and finding six was missing.

1. Sovereignty is a control system.	Residency and local inference can help, but they do not by themselves decide who holds keys, sets policy, approves actions, sees evidence, or can replace a supplier.
2. The home needs a shared map, not one universal memory.	People, roles, accounts, devices, documents, tasks, and permissions need explicit relationships. Personal, child, guest, shared, and employer knowledge remain separate by default. Task-specific capability matters more than a monolithic AI label. [4]
3. Replacing the model is now the easy part.	Platform software has taken this mechanism. An operating-system framework now accepts any model provider behind one application interface, and the common tool protocol renders supplier interfaces on the host's own consent and audit path. [17][21][22] What substitution does not carry is the interesting residue: replaying the evidence that a capability still behaves the same, keeping authority bound to it, preserving a recorded objection, and leaving a contestable receipt. The task boundary, context contract, quality bar, approval rule, and recovery path should hold across the swap. The swap itself is no longer the claim.
4. Authority belongs to typed actions.	Fluent output is not permission. Explain, draft, pay, order, unlock, share, and delete are distinct actions with distinct identity, state, policy, approval, and evidence requirements. Bounded delegation now has a shipping precedent in signed payment mandates, which separate a concrete human-present approval from a bounded human-not-present intent. [23]
5. Exit, continuity, and evidence are the test.	A sovereign household can keep safe functions working through disruption, substitute approved suppliers, export its map and policies, and show what context, route, decision, and change occurred.
6. Someone is always overruled. Record it.	A household is not one user and a quorum is not a consensus. A guardian can be outvoted, a teenager may object to a grant made over them, a caregiver and an adult child may disagree about an older parent's home. A system that stores only the decision has discarded the objection. A published gap analysis of the current agent protocols finds dissent preservation absent from every one surveyed, so the record of who disagreed, what it did to the grant, and how to revisit it belongs in the household layer or nowhere. [24]



WHAT IS ALREADY IN FORCE

The clock started while we were writing.

This argument used to need a reason to act now. It has one. Two European obligations came into force after the first edition, and both reach the ordinary household products described here. [27][28][29] A vendor will produce disclosure, oversight, provenance, and incident evidence regardless; the claim is that producing it once, as a portable capability contract, is cheaper than producing it per product, per model, per market.

Date	Status	Obligation	Carried by
2 Aug 2026	In force	EU AI Act, Article 50 People must be told when they are interacting with an AI system, and synthetic content must be marked.	Context grant, action receipt
11 Sep 2026	In force	Cyber Resilience Act, Article 14 Actively exploited vulnerabilities and severe incidents reported within 24 hours, notified within 72, closed within 14 days, including products already placed on the market.	Capability lease, evidence duty
2 Dec 2026	Next	Article 50 marking grace ends The narrower marking duty reaches systems already deployed.	Action receipt
2 Dec 2027	Deferred	AI Act high-risk regime, Annex III Deferred sixteen months from August 2026. Conformity documentation, risk management, and human oversight for stand-alone high-risk systems.	Conformance, authority, dissent
11 Dec 2027	Upcoming	Cyber Resilience Act, full conformity CE marking and conformity assessment for products with digital elements.	Capability lease, provenance
2 Aug 2028	Upcoming	AI Act, Annex I embedded High-risk duties for AI built into already-regulated products.	Conformance, provenance

A mapping, not advice

This is a design mapping, not legal advice, and not a claim that any household capability is classified high-risk - classification depends on how a system is deployed. Dates are obligations on providers and manufacturers placing products on the EU market. Non-EU regimes are outside the scope of this brief.

THE HOUSEHOLD OPERATING GRAPH

Shared meaning. Separate knowledge.

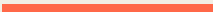
The architecture needs a policy-filtered map of household reality, not a centralized copy of every private record. It stores identities, relationships, permissions, pointers, minimum shared state, and evidence while leaving private content in its authorized scope where possible.

People and roles	Adult, guardian, child, guest, caregiver, delegated operator, employer identity.
Places and devices	Rooms, phones, hubs, PCs, appliances, vehicles, engines, trust and resource state.
Accounts, assets, obligations	Bills, subscriptions, warranties, appointments, inventory, vehicles, service relationships.
Knowledge scopes	Private, shared, guardian, child, guest, and employer records with provenance and retention.
Capabilities	Rules, specialists, generalists, remote providers, validators, tools, versions, and evaluated scope.
Policies	Purpose, consent, context grants, authority tier, residency, budget, deadline, escalation.
Typed actions	Parameters, preconditions, permissions, effects, reversibility, approval, and evidence.
Evidence and outcomes	Routes, grants, validations, approvals, receipts, postconditions, errors, and undo windows.

The graph is not a digital twin.

Household state can be incomplete, stale, disputed, or inferred. Every assertion needs source, owner, freshness, sensitivity, and policy. Missing information is not proof that something is false.

Open, machine-readable device descriptions, typed actions, and provenance models offer useful standards precedents for this layer. [8][9]



THE CONTROL LOOP

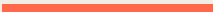
Every request crosses five gates.

The household states a job. The system resolves the applicable people, assets, relationships, context, and current state; enforces authority before optimization; chooses and verifies a capable implementation; then acts, asks, hands off, or stops.

01 Name What job needs to be finished?	02 Resolve Which people, assets, and facts apply?	03 Authorize Which context and actions are allowed?	04 Run Choose and verify a proven capability.	05 Act Execute, ask, hand off, or stop.
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Context grant	Minimum authorized data references, purpose, fields, expiry, retention, and policy version.
Capability lease	Approved function, model or rule; allowed device; tools; resource fit; timeout; fallback.
Action proposal	Typed target, parameters, expected state, risk, reversibility, approval, and idempotency.
Action receipt	Requester, context, route, versions, policy decision, validation, approval, effects, and outcome.

The model may propose. The action contract decides whether anything happens.	Consequential changes run only through explicit action interfaces. Identity, purpose, target state, parameters, policy, and approval are rechecked at commit time. An intelligible denial, abstention, deferral, or handoff is a valid outcome. [5][6][7][20][23]
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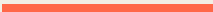


CONCRETE HOUSEHOLD EXAMPLES

Useful because authority is legible.

Catch a bill increase Resolve the account owner, tariff, prior statements, requester role, and evidence sources before a model sees the task. Boundary: Explain and draft. Do not pay, cancel, or close an account without fresh approval.	Help a child learn Resolve the child, assignment, guardian rules, and only the school context approved for the learning task. Boundary: Teach and check understanding. Do not complete assessed work or impersonate the student.
Troubleshoot an appliance Resolve the exact appliance, warranty, service history, and manual before a specialist proposes a check. Boundary: Guide low-risk, manual-supported steps. Stop for hazards or warranty-voiding work.	Plan groceries Combine only authorized inventory, shared lists, allergies, and preferences without flattening private profiles. Boundary: Suggest and update the shared list. Do not merge private data or check out without approval.

Household activity surveys can identify recurring task families, but they are demand evidence - not proof that a task should be automated. [10] These examples are design tests, not claims of empirical performance. Smart-home task competence is already well covered by public benchmarks; what none of them measures is whether these guarantees survive an implementation change. [32]



IMPLICATIONS

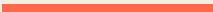
The operating layer becomes strategic.

Builders Build the household operating graph. Represent people, roles, assets, context scopes, capabilities, policies, actions, and evidence. Bind models only after the outcome and authority contract exists.	Device makers Expose capability, trust, and action. Publish typed device capabilities, action interfaces, trust boundaries, and measured resource envelopes under real operating constraints.	Investors Test whether the moat respects exit. Governed household state, permission relationships, typed actions, and outcome evidence matter only if they remain portable and auditable.
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Seven tests, plus economic reality

Authority	Can policy permit, constrain, approve, or deny the action?
Context	Do personal, shared, child, guest, and employer scopes stay separate?
Substitutability	Can a supplier change without losing capability, policy, or history?
Continuity	Is there safe degraded operation, recovery, export, and exit?
Evidence	Can the household see what context, route, decision, and change occurred?
Dissent	Is a recorded objection preserved, visible in the receipt, and revisitable?
Economic reality	Do benefits survive full operating, validation, and failure costs?

Economic rule: optimize for cost per policy-compliant successful outcome, not token price. Count storage, updates, energy, cold starts, device interference, validation, routing errors, human review, outage recovery, portability work, and failure loss. Model-routing research suggests potential cost-quality gains; the household thesis adds authority, device, continuity, dissent, and evidence constraints. [13][14][15]



WHAT WAS BUILT AND MEASURED

The contract runs. So does the metric.

Every prior edition proposed a capability contract and an evaluation agenda, and measured nothing. Since the second edition the contract has been built as a machine-readable schema with three validating reference capabilities, and the substitution metric has been computed against it. This is the first measured figure in this work.

Nine of the twelve conformance gates turn out to need no model at all. They are static analysis of a contract — the authority ceiling, the context scope, the shape of the receipt, whether every declared state can still reach an ending. That is what this brief’s own policy-before-placement invariant predicts, and it is the cheaper half of the program. The remaining three need a running implementation and report as not evaluated rather than being quietly dropped. Ten deliberately invalid contracts, each naming the gate it must break, keep the runner honest: it fails if any of them passes.

The substitution result

Measure	Value	What it means
Outcome, replacement	0.955	Follows a merchant through a rename; better at the job
Outcome, incumbent	0.809	The careful binding it would replace
Aggregate preservation	0.947	Reads as an acceptable substitution
Worst component	0.859	Stops disclosing when the task crosses to a remote service

A binding that is better at the task and should still not ship. The aggregate barely moves because an outcome gain pays for a policy loss; the worst-component floor is what catches it. That rule was specified in an earlier edition on the argument that averaging hides exactly this, and it had never been demonstrated. It has now. Thirty-seven authored scenarios run across twenty ordered substitutions, both directions of every swap, because stability is not symmetric.

The harness also had to be tested against itself. A first cut scored a prohibited-action check that no implementation could ever trigger, so five adversarial scenarios passed trivially and read as evidence. The scorer now names every enforcement check, counts how often each is violated, and refuses to report a figure if any check has never run at all.

What the figure does not earn

These are deterministic implementations, not models, so this measures whether the contract is specific enough to catch a regression — not what happens when one model is swapped for another. The scenarios are authored fixtures, not field data. The stakes weighting is chosen, not derived. Three of the twelve gates remain unevaluated. The judgment is unchanged.

PRIOR ART, BOUNDARIES, FALSIFICATION

The integration is the test.

Fog and service-routing work already address heterogeneous placement. Model routers trade quality and price. Consumer frameworks and smart-home pipelines already choose runtimes and compose services. Privacy systems use attestation and minimized cloud processing. Governance and web standards already provide lifecycle controls, typed device interactions, and provenance. Agent protocols now supply tool transport, discovery, and host-rendered interfaces. [6][8][9][11][12][13][14][15][16][17][18][19][21][22][25][26]

Two claims in the first edition were narrower than they read. The stability of a product across a model change restates an established machine-learning result on backward compatibility, where a regression on a previously-passing case is a negative flip; the household contribution is raising that measurement from a prediction to a whole contract. [30] And the argument that governance is a missing layer above the agent protocols was published independently before this brief was revised; the residue here is domestic and multi-principal rather than enterprise. [24]

Policy before optimization	No cheaper, faster, or more capable route may weaken context, identity, jurisdiction, or authority.
Competence before placement	No model or bundle serves a task family until the complete workflow passes its evaluation gate.
No direct model writes	Consequential changes use typed actions with admission, current-state checks, approval, and receipts.
Feedback cannot rewrite policy	Verified outcomes may propose changes; evaluation, approval, canary, and rollback govern adoption.

The honest falsification test

The thesis loses if one compact generalist plus tools delivers the same outcomes, control, continuity, portability, and full cost with less complexity. It also loses if graph maintenance becomes too costly or stale, competence boundaries are unreliable, device fragmentation overwhelms gains, or supplier substitution is performative. One condition is now partly met: the platform absorbed substitution itself. What it has not absorbed is conformance replay, bound authority, recorded dissent, and a contestable receipt — the part this edition built and measured. If those are absorbed too, the category does not survive.

Judgment: useful synthesis but not yet novel. A reference implementation and a reproducible benchmark now exist, which every prior edition lacked. What they do not yet show is these contracts outperforming simpler baselines under a real model substitution, and until a model binding runs against the same harness the judgment does not move.

Evaluation caveat: simulated households are useful for generating hypotheses and finding crashes, and unreliable as evidence. Published work finds agent success varies by up to nine points depending on which model plays the user, with calibration errors falling hardest on dialect and age groups a household product must serve. Hard gates need human-subject or recorded-interaction evidence, reported stratified. [31]



SELECTED SOURCES

Evidence behind the brief

Sources prioritize public frameworks, official standards, first-party technical documentation, and peer-reviewed or clearly labeled preprint research. Entries [1] to [20] are carried from the first edition with their original numbers so inline citations remain stable; [21] to [32] are new in this edition. Accessed September 2026.

- [1] European Commission. Cloud Sovereignty Framework: Implementation Guidance (June 2026).
- [2] Stanford HAI. 2026 AI Index, Chapter 8: Policy and Governance, section 8.3.
- [3] Andrea Renda et al. Is AI Sovereignty Possible? Brookings / CEPS (February 2026).
- [4] Brookings. The Myth of the Monolith: AI Is Not One Thing (October 2025).
- [5] Singapore IMDA. Model AI Governance Framework for Agentic AI (2026).
- [6] NIST. AI Risk Management Framework Core. <https://airc.nist.gov/airmf-resources/airmf/5-sec-core/>
- [7] Brookings / Carnegie Mellon University. How Can We Best Evaluate Agentic AI? (April 2026).
- [8] W3C. Web of Things Thing Description 1.1 Recommendation (December 2023). <https://www.w3.org/TR/wot-thing-description11/>
- [9] W3C. PROV-O: The PROV Ontology. <https://www.w3.org/TR/prov-o/>
- [10] U.S. Bureau of Labor Statistics. American Time Use Survey and Consumer Expenditure Surveys.
- [11] NIST. SP 500-325: Fog Computing Conceptual Model. <https://csrc.nist.gov/pubs/sp/500/325/final>
- [12] IETF. Computing-Aware Traffic Steering (CATS). <https://datatracker.ietf.org/group/cats/about/>
- [13] Lingjiao Chen, Matei Zaharia, James Zou. FrugalGPT (preprint, 2023). <https://arxiv.org/abs/2305.05176>
- [14] LMSYS Org. RouteLLM (preprint, 2024). <https://arxiv.org/abs/2406.18665>
- [15] Francisco Romero et al. INFaaS (preprint, 2019). <https://arxiv.org/abs/1905.13348>
- [16] Apple. Private Cloud Compute Security Guide. <https://security.apple.com/documentation/private-cloud-compute/>
- [17] Apple Developer. Foundation Models framework. <https://developer.apple.com/documentation/foundationmodels/>
- [18] Home Assistant Developers. Voice pipelines. <https://developers.home-assistant.io/docs/voice/pipelines/>
- [19] Microsoft Learn. Windows ML overview. <https://learn.microsoft.com/en-gb/windows/ai/new-windows-ml/overview>
- [20] Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. NeurIPS 2017.
- [21] Apple. Foundation Models framework opened to third-party model providers, WWDC 2026 (June 2026).
- [22] Model Context Protocol. Specification revision 2026-07-28: stateless core, extensions framework, feature lifecycle policy. <https://modelcontextprotocol.io/specification/2026-07-28>
- [23] Google / FIDO Alliance. Agent Payments Protocol (AP2): Intent and Cart Mandates as signed verifiable credentials. <https://ap2-protocol.org/ap2/specification/>
- [24] Governance Gaps in Agent Interoperability Protocols: What MCP, A2A, and ACP Cannot Express (preprint, June 2026). <https://arxiv.org/abs/2606.31498>
- [25] Model Context Protocol. MCP Apps: host-rendered interfaces on the shared consent path (January 2026). <https://modelcontextprotocol.io/extensions/apps/overview>
- [26] A2A Project / Linux Foundation. A2A Protocol v1.0 (April 2026); v1.0.1 extension mechanism (May 2026). <https://a2a-protocol.org/latest/specification/>
- [27] European Union. Regulation (EU) 2024/1689, Article 50: transparency obligations, applicable 2 August 2026. <https://artificialintelligenceact.eu/article/50/>
- [28] European Union. Digital Omnibus on AI: deferral of Annex III high-risk obligations to 2 December 2027 (Council approval, June 2026).
- [29] European Union. Regulation (EU) 2024/2847 (Cyber Resilience Act), Article 14: reporting obligations applicable 11 September 2026. <https://eur-lex.europa.eu/eli/reg/2024/2847/oj>
- [30] Srivastava, Nushi et al. An Empirical Analysis of Backward Compatibility in Machine Learning Systems. KDD 2020; and A Generalized Backward Compatibility Metric, KDD 2022.
- [31] Seshadri et al. Lost in Simulation: LLM-Simulated Users are Unreliable Proxies for Human Users in Agentic Evaluations. ACL 2026. <https://aclanthology.org/2026.acl-long.2192/>
- [32] HomeBench: Evaluating LLMs in Smart Homes with Valid and Invalid Instructions Across Single and Multiple Devices. ACL 2025. <https://aclanthology.org/2025.acl-long.597/>

Scope note: this brief proposes an architecture and evaluation agenda. It does not claim that the operating graph is complete, that typed actions make arbitrary agents safe, that external effects are always reversible, or that model substitution works without system-level regression testing. One figure in this brief is now measured — the substitution result above, which covers deterministic implementation substitution over authored fixtures and not a model

swap. Every other figure remains a scenario assumption or a design target.

